

Chapter 2 Matrix Algebra

- **2.1 Matrix operations:** be able to compute the basic operations and know when the operation is undefined.
- **2.2 Linear transformations:**
 - Find the standard matrix for a given a given linear transformation, e.g., HW 2.
 - * Find the columns of the standard matrix by using the column vectors from the standard basis., e.g., $T\left(\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}\right)$, $T\left(\begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}\right)$, and $T\left(\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}\right)$ gives the 1st, 2nd, and 3rd columns respectively.
 - Determine if a given function is a *linear transformation*, e.g., HW 4ab.
 - * Recall that to prove that T is a linear transformation it suffices to find a standard matrix, $A \in \mathcal{M}_{m \times n}$ for $\mu_A : \mathbb{R}^n \rightarrow \mathbb{R}^m$. To prove that it is not a standard matrix you need to show that it fails at least one of the properties.
- **2.3 Inverse matrices.**
 - Use Gaussian elimination to determine if a matrix A is invertible and find A^{-1} .
 - Use the inverse A^{-1} to solve $A\mathbf{x} = \mathbf{b}$ e.g., HW 2ab.

Theorem. *An $n \times n$ matrix is nonsingular $\iff$ it is invertible.*

Chapter 3 Vector Spaces

- **3.1 Subspaces of $\mathbb{R}^n$.**
 - Determine if a given set is a *subspace*, e.g. HW 1ab.
- Proposition.** *Let $A \in \mathcal{M}_{m \times n}$. The set of solutions to $A\mathbf{x} = \mathbf{0}$ is a subspace of $\mathbb{R}^n$*
- Proposition.** *Let $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k \in \mathbb{R}^n$. Then $V = \text{Span}(\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k)$ is a subspace of $\mathbb{R}^n$.*
- Decide whether a given set of vectors *spans* $\mathbb{R}^n$, e.g., HW 2ab.
 - * Note: you can use dimensions (section 3.4) to determine that some sets do not span $\mathbb{R}^n$.
 - ★ Let $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k \in \mathbb{R}^n$ and let $\mathbf{v} \in \mathbb{R}^n$. Prove that $\mathbf{v} \in \text{Span}(\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k) \Rightarrow \text{Span}(\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k) = \text{Span}(\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k, \mathbf{v})$.
- **3.3 Linear Independence and Basis**
- Proposition.** *Let $A \in \mathcal{M}_{n \times n}$. Then A is nonsingular if and only if its column vectors form a basis for $\mathbb{R}^n$.*
- Determine if a given set of vectors is *linearly independent*, e.g., HW 2ab.
 - * Find the reduced echelon form of the matrix of column vectors; $A\mathbf{x} = \mathbf{0}$ has only the trivial solution $\Rightarrow$ the vectors are linearly independent.
 - Determine if a given set of vectors is a *basis* for a given space, e.g., HW 3bc
 - * To show that $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k \in \mathbb{R}^n$ is a basis, you must show that $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k\}$ are linearly independent (see above) and $\text{Span}\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k\} = \mathbb{R}^n$. If the set fails either of these, it is not a basis for $\mathbb{R}^n$.
 - * Section 3.4 can be used to show that a set is not a basis.

• **3.4 Dimension and Its Consequences.**

Proposition. Let $V \subset \mathbb{R}^n$ be a subspace, let $\{\mathbf{v}_1, \dots, \mathbf{v}_k\}$ be a basis for V , and let $\{\mathbf{w}_1, \dots, \mathbf{w}_\ell\} \in V$. If $\ell > k$ then $\{\mathbf{w}_1, \dots, \mathbf{w}_\ell\}$ must be linearly dependent.

- $\dim \mathbb{R}^n = n$; meaning that a basis for $\mathbb{R}^n$ such as the standard basis will have n vectors.
- Any two bases for a space will have the same number of vectors.
- Let $\{\mathbf{v}_1, \dots, \mathbf{v}_k\} \subset \mathbb{R}^n$. If $k \neq n$ then $\{\mathbf{v}_1, \dots, \mathbf{v}_k\}$ is not a basis for $\mathbb{R}^n$. If $k = n$ then it is a basis if and only if $\{\mathbf{v}_1, \dots, \mathbf{v}_k\}$ linearly independent.

Example. (HW 1.b) Find a basis for each of the given subspaces and determine its dimension.

1. $V = \{\mathbf{x} \in \mathbb{R}^n : x_1 + x_2 + x_3 + x_4 = 0, x_2 + x_4 = 0\} \subset \mathbb{R}^4$.

$$\left[\begin{array}{cccc|c} 1 & 1 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 & 0 \end{array} \right] \rightsquigarrow \left[\begin{array}{cccc|c} 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 \end{array} \right] \Rightarrow x_1 = -x_3 \text{ and } x_2 = -x_4$$

$\Rightarrow$ the general solution is $\mathbf{x} = x_3(-1, 0, 1, 0) + x_4(0, -1, 0, 1)$. So V is spanned by $(-1, 0, 1, 0)$ and $(0, -1, 0, 1)$. It's easily checked that these vectors are linearly independent. So V is a basis with $\dim V = 2$.

2. $V = \text{span}\{(3, -2, 4), (-3, 2, -4), (-6, 4, -8)\}$.

$$\left[\begin{array}{ccc} 3 & -3 & -6 \\ -2 & 2 & 4 \\ 4 & -4 & -8 \end{array} \right] \rightsquigarrow \left[\begin{array}{ccc} 1 & -1 & -2 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{array} \right]$$

So $(3, -2, 4)$ (as is the other vectors) is a basis and $\dim V = 1$.